34. Develop a semantic search system using cosine similarity between document and query embeddings.

Code:

```
from sentence_transformers import SentenceTransformer
from sklearn.metrics.pairwise import cosine_similarity

# Load embedding model
model = SentenceTransformer("all-MiniLM-L6-v2")

# Documents
documents = [
    "Python is a popular programming language.",
    "Machine learning allows computers to learn from data.",
    "Artificial intelligence is used to build smart systems.",
    "Natural language processing deals with human language.",
    "Python is widely used in data science and machine learning."
]

# Get query from user
query = input("Enter your search query: ")

# Generate document embeddings
document_embeddings = model.encode(documents)

# Generate query embedding
query_embedding = model.encode([query])

# Calculate cosine similarity
```

```
scores = cosine_similarity(
    query_embedding,
    document_embeddings
)[0]

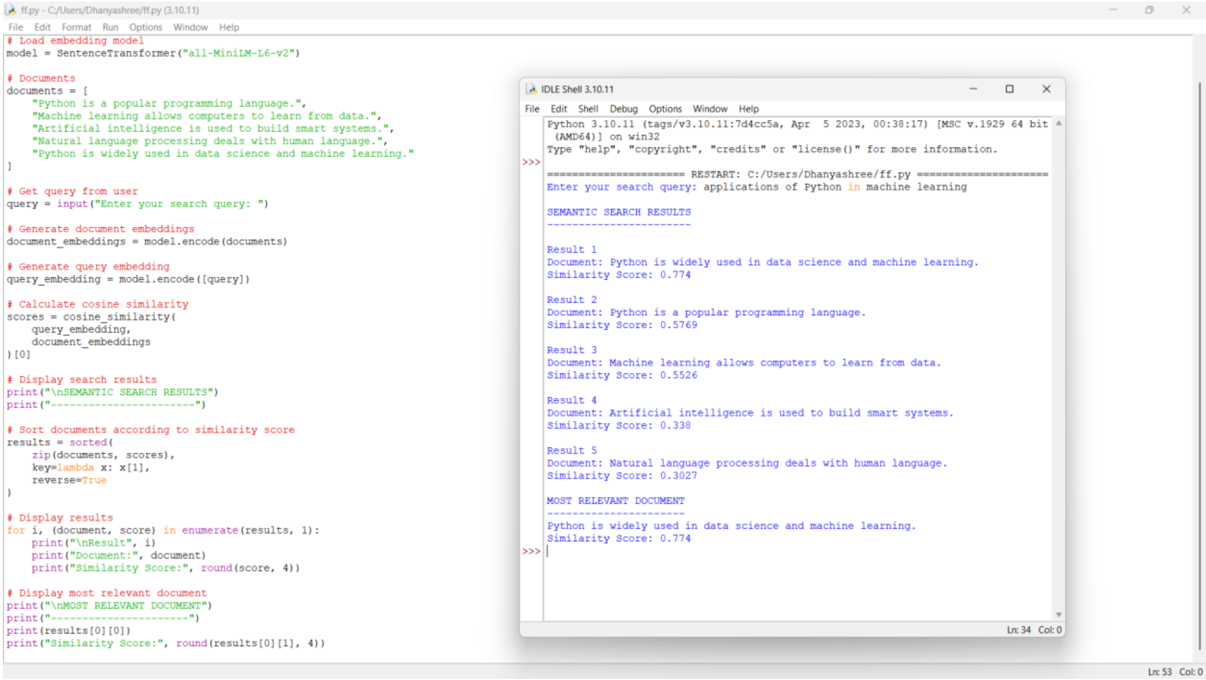
# Display search results
print("\nSEMANTIC SEARCH RESULTS")
print("-----------------------")

# Sort documents according to similarity score
results = sorted(
    zip(documents, scores),
    key=lambda x: x[1],
    reverse=True
)

# Display results
for i, (document, score) in enumerate(results, 1):
    print("\nResult", i)
    print("Document:", document)
    print("Similarity Score:", round(score, 4))

# Display most relevant document
print("\nMOST RELEVANT DOCUMENT")
print("----------------------")
print(results[0][0])
print("Similarity Score:", round(results[0][1], 4))
```

output:



ff.py - C:/Users/Dhanyashree/ff.py (3.10.11)

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```

Ln: 53 Col: 0

IDLE Shell 3.10.11

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```
Python 3.10.11 (tags/v3.10.11:7d4cc5a, Apr  5 2023, 00:38:17) [MSC v.1929 64 bit
 (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
====================== RESTART: C:/Users/Dhanyashree/ff.py =====================
Enter your search query: applications of Python in machine learning

SEMANTIC SEARCH RESULTS
----------------------

Result 1
Document: Python is widely used in data science and machine learning.
Similarity Score: 0.774

Result 2
Document: Python is a popular programming language.
Similarity Score: 0.5769

Result 3
Document: Machine learning allows computers to learn from data.
Similarity Score: 0.5526

Result 4
Document: Artificial intelligence is used to build smart systems.
Similarity Score: 0.338

Result 5
Document: Natural language processing deals with human language.
Similarity Score: 0.3027

MOST RELEVANT DOCUMENT
---------------------
Python is widely used in data science and machine learning.
Similarity Score: 0.774
>>>
```

Ln: 34 Col: 0